Paleomagnetism of the Miocene Grotto and Snoqualmie Batholiths, Central Cascades, Washington

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Paleomagnetic data have been obtained from granitic rocks of the Miocene Grotto and Snoqualmie batholiths in the Central Cascades, Washington. Alternating field magnetic cleaning demonstrated that a wide range of magnetic stability was present and that many specimens were unsuitable for paleomagnetic analysis. A simple statistic (Q_d'), defined as the ratio of remanent magnetic intensity of the entire sample after af cleaning to weak-field magnetic susceptibility, proved effective in discriminating between specimens of marginal stability. Experiments with saturation isothermal remanent magnetism indicated that stable NRM is dominated by magnetic grains with single-domain characteristics, whereas grains with multidomain characteristics dominate the NRM of the unstable samples. After cleaning, a Miocene pole was obtained for the Central Cascades at 208.5°E, 85.5°N ($\delta_p = 4.5$, $\delta_m = 5.4$, and $k = 360.4$). This pole is statistically indistinguishable from other Miocene poles of North America, which suggests that any major tectonic movement postulated for the Central Cascades relative to North America must have occurred before the Miocene.

In a recent study of Cretaceous Central Cascade intrusives *Beck and Noson* [1972] interpreted their paleomagnetic results to suggest that west central Washington was at anomalously low paleolatitudes during the Cretaceous. Their data indicate that the Cretaceous part of the Central Cascade area was approximately 30° farther south than cratonic North America. Tectonic translation has also been postulated by *Monger and Ross* [1971] in one of their proposed models of the western Cordillera based on the distribution of fusulinaceans. Other earlier theories such as *Wise's* [1963] application of Carey's orocline concept to the Pacific Northwest require tectonic rotation of the Central Cascades. Paleomagnetic data from Miocene granitic intrusions summarized in this paper suggest that the Central Cascades have not undergone any post-Miocene tectonic rotation or translation. Therefore any tectonic movements required by these theories must have ended by the Miocene. In addition, preliminary work on an upper Eocene granitic stock shows similar results and suggests that these observations might be pushed back even further into the Tertiary.

GEOLOGICAL SETTING

The Miocene Grotto and Snoqualmie batholiths are exposed just west of the crest of the Cascade Range, 64 km east of Seattle (Figure 1). These intrusions are composed predominantly of granodiorite and contain lesser amounts of diorite, quartz diorite, and quartz monzonite. There is no clear petrologic distinction between samples from the two batholiths, and they were thought to be the same unit until radiometric age dating revealed their separate histories [*Erickson*, 1968]. The K-Ar ratios from biotite and hornblende date the Grotto batholith at 26.3 ± 0.8 and 25.0 ± 1.5 m.y. [*Yeats*, 1970]. The K-Ar ratios on biotites from the Snoqualmie batholith yield an age of 18 m.y. [*Baadsgaard et al.*, 1961] and 17 ± 0.7 m.y. [*Curtis et al.*, 1961], whereas an apatite fission track date gives an age of 15.5 ± 1.5 m.y. for the Snoqualmie batholith. The Grotto and Snoqualmie batholiths were mapped and described by *Yeats* [1958] and *Erikson* [1968, 1969].

Misch [1966] described the complex geology

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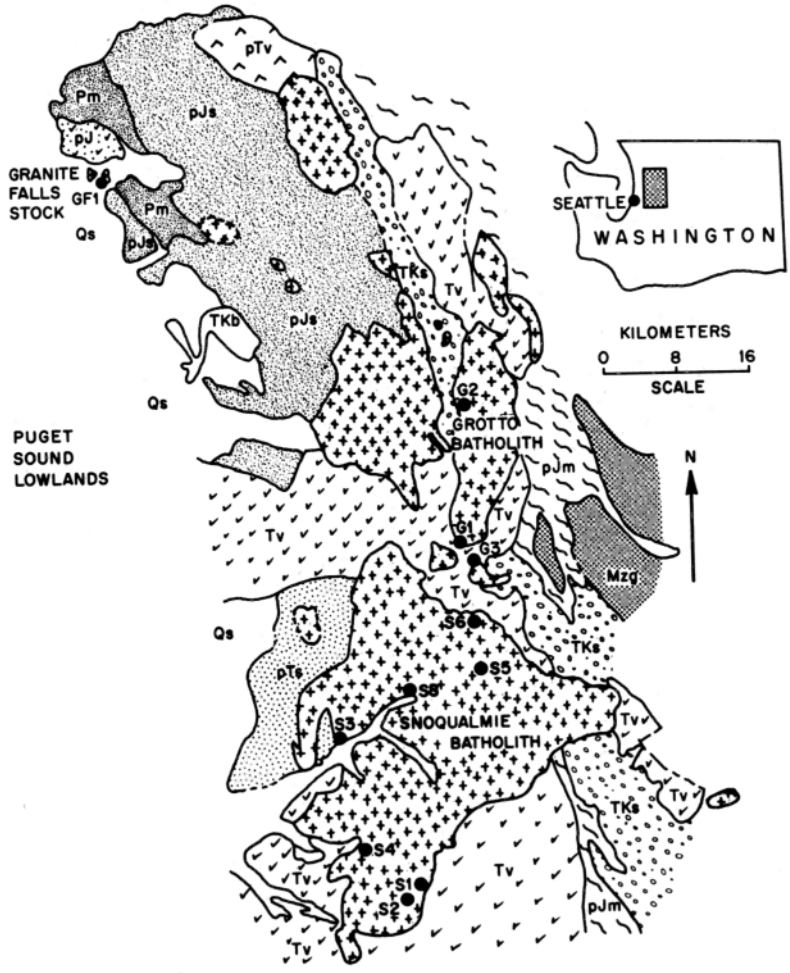


Fig. 1. Geologic and index map of Central Cascades, Washington. Geology modified from *Huntting et al.* [1961]. Qs, Quaternary sediments; plus signs, Tertiary intrusives; Tv, Tertiary volcanics; TKs, Upper Cretaceous Paleocene sediments; Mzg, Cretaceous intrusives; pJs, pre-Jurassic sediment; pJm, pre-Jurassic metamorphics (phyllites, greenschists, schists, gneiss); Pm, Permian metasediments.

of the Central and Northern Cascades, which is briefly summarized below. Pre-Jurassic rocks in the Central Cascades consist of metamorphic equivalents of oceanic suites with complex structures that trend northwest-southeast [*Misch*, 1966]. During the Middle Cretaceous large sheets of slightly metamorphosed Paleozoic volcanic and sedimentary rocks were thrust over late Permian(?) metamorphics. After thrusting, the granitic core of the present Cascades was emplaced in two distinct episodes. The older granitic complex includes the 94- to 83-m.y. Mt. Stuart batholith and Beckler Peak stock. Paleomagnetic data from these plutons have been summarized in a preliminary report by *Beck and Noson* [1972]. The younger granitic intrusions were formed 48–13 m.y. ago and are capped by andesitic volcanic rocks erupted during the same time period. Age dates suggest that these granitic rocks were added to the core of the present Cascades during much of the Cenozoic [*Yeats*, 1970]. The present Cascades were formed by uparching in the late Pliocene.

METHODS AND RESULTS

Oriented samples were collected by using a standard paleomagnetic field drill similar to that described by *Doell and Cox* [1965] and

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were oriented in situ with a magnetic compass. Back azimuths were taken to ensure against orientation error caused by local magnetic anomalies. Remanent magnetism was measured with a Schönstedt spinner magnetometer, and demagnetization experiments were performed by using an alternating magnetic field and a three-axis tumbler [*Doell and Cox*, 1967]. Initially, two or more pilot specimens from each site were chosen for progressive af demagnetization. The cleaning level at which the pilot specimens reached their final stable magnetization directions was chosen as the optimum cleaning level for the entire site.

Early in the magnetic cleaning process it became obvious that a very wide spectrum of magnetic stability was present in our samples. At some sites the degree of magnetic stability was uniform, whereas at others it varied markedly from sample to sample over a distance of a few tens of meters. No simple correlation of magnetic stability with obvious weathering or secondary alteration has been recognized. In order to reduce the spurious dispersion that would be introduced by including data from magnetically unstable specimens or sites, it was necessary to conduct fairly elaborate stability tests and to reject specimens judged to be unstable. Most of the laboratory work was concerned with this problem. The paragraphs that follow summarize our methods and results. For a more complete discussion of the stability problem, pertaining specifically to the Snoqualmie batholith, see *Beske* [1972].

In order to study magnetic stability, specimens were subjected to af magnetic cleaning at 150-, 200-, and 250-oe peak field intensities. Each specimen was demagnetized twice, at each level, once with its $+z$ axis oriented into the innermost tumbler and once with the opposite orientation.

Specimens were next classified as stable, marginally stable, or unstable on the basis of their reaction to the magnetic cleaning process. Unstable samples showed the following characteristics. (1) Their magnetic intensities dropped very abruptly in response to magnetic cleaning at low af intensities. (2) No final stable direction was reached, even when magnetic cleaning was extended to fairly high fields (of the order of 1000 oe). (3) Repeated magnetic cleaning and measurement at the same level yielded directions that differed by 15° or more. (4) Their directions for the most part diverged by considerable angles (30° or more) from what appeared to be the mean direction for the site. Obviously stable samples showed the reverse of all four characteristics. The stability was also amply demonstrated by the antiparallelism of the normal and reversed directions after af demagnetization. Slightly more than half of our specimens could be classified unequivocally as stable or unstable by these criteria.

A less subjective criterion was needed to deal with marginally stable samples. After experimenting with several published stability tests, we concluded that the statistic with the greatest ability to discriminate between stable and unstable specimens in our granitic rocks was Q_d', which we define as the ratio of intensity of remanent magnetization of the entire specimen after af demagnetization at the optimum field (see above) to low-field magnetic susceptibility. For sites so unstable that no optimum field strength for magnetic cleaning could be determined, the magnetic intensity after demagnetization at 150 oe was used in calculating Q_d'.

As is seen in Figure 2, it is in the range from 0.19 to 0.22 of Q_d' values where the angular difference ($\Delta°$) between the two directions of magnetism for a sample at the same cleaning level increased remarkably. In our collection of

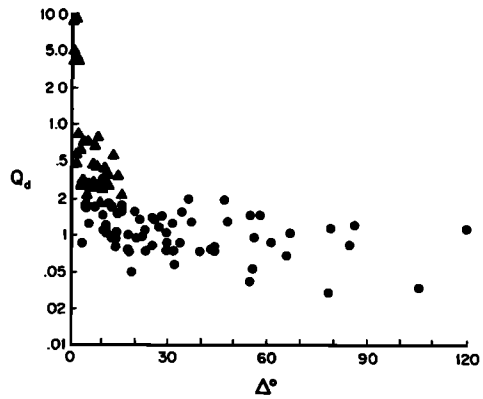


Fig. 2. Correlation between the stability ratio Q_d' and the angular difference ($\Delta°$) between two directions of remanent magnetization obtained after repeated af magnetic cleaning at the same peak field value. Samples judged as stable (unstable) by stability criteria other than criterion Q_d' are indicated by triangles (circles).

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TABLE 1. Summary of Paleomagnetic Data from Tertiary Intrusions, Central Cascades, Washington

Site	No. of Samples*	af Cleaning Level, oe	Declination, °E	Inclination	Length of Resultant R	Precision Parameter k	95% Confidence, deg	Paleomagnetic Pole
				Snoqualmie Batholith†				
S1	5 (16)	200	170.0	-69.0	4.99	291.5	4.5	81.5°N, 190.5°E
S3	7 (12)	150	165.0	-70.5	6.94	106.8	5.8	78.0°N, 190.5°E
S4	12 (5)	200	13.5	69.5	11.94	178.6	3.3	79.5°N, 290.5°E
S5†	6§ (11)	200	128.5	-75.5	5.90	48.5	9.7	58.0°N, 196.0°E
S6	10 (11)	150	360.0	66.5	9.69	28.8	9.2	88.5°N, 234.5°E
				Grotto Batholith				
G1	10 (10)	300	4.0	66.5	9.60	22.6	10.4	87.0°N, 304.0°E
G2	12 (12)	100	167.5	-65.0	11.97	333.8	2.4	81.5°N, 146.5°E
G3	9 (9)	300	355.5	69.5	8.92	101.1	5.1	84.0°N, 212.0°E
				Granite Falls Stock				
GF1	5 (5)	150	182.0	-67.5	4.98	205.8	5.4	87.5°N, 272.0°E
Cascade Miocene pole	7 (10)		365.5	68.5	6.98	360.5	3.2	85.5°N, 208.5°E

δ_p = 4.5; δ_m = 5.4. Snoqualmie batholith, 15 to 18 m.y.; Grotto batholith, 23.5 to 27.0 m.y.; Granite Falls stock, 43.3 ± 1.4 m.y.
*Number of samples used in site mean calculations; number in parentheses is the number of samples collected.
†Sites S2, S5, S7, and S8 were not included in Miocene paleomagnetic pole calculation. See text for explanation.
§Number of samples before application of Q_d' criteria.

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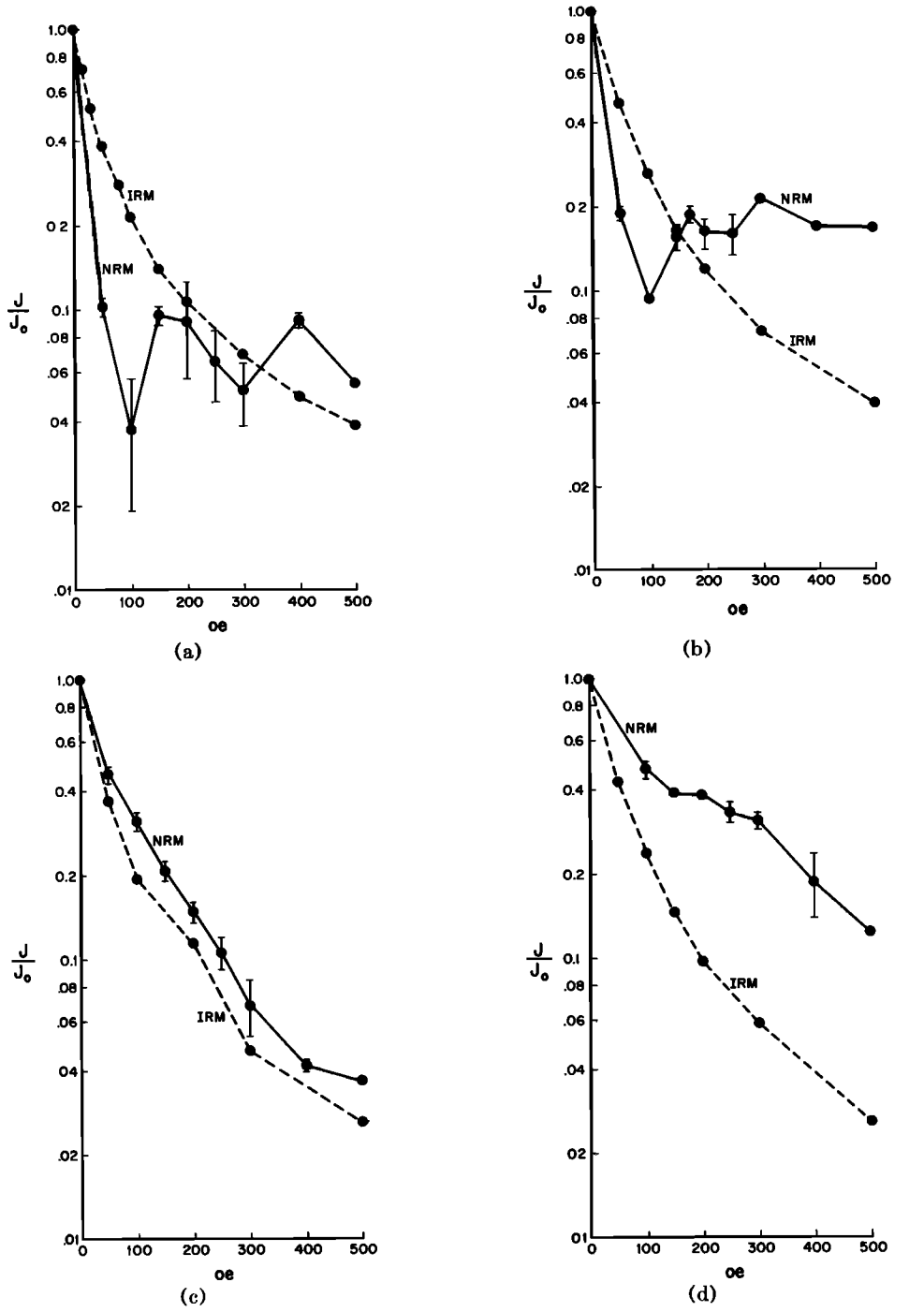


Fig. 3. Semilog plot comparing the stability of the natural remanent magnetism NRM and the saturation isothermal remanent magnetism IRM to af magnetic cleaning of representative specimens from the Snoqualmie batholith. Here J is the intensity of magnetism at the particular cleaning level, and J_0 is the intensity of NRM before cleaning. (*a*) Specimen 1B037A, unstable. (*b*) Specimen 1B353, unstable. (*c*) Specimen 1B061B, stable. (*d*) Specimen 1B328, stable.

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Tertiary granitic rocks from Washington, stable samples are characterized by values above this range (greater than 0.22), whereas unstable samples have Q_d' values below this range (less than 0.19). Specimens between these values were classified as stable or unstable by using the subjective criteria mentioned above. Use of the Q_d' ratio was effective in reducing dispersion very markedly at several sites. The Q_d' criterion also indicated that all but two of the specimens from site S5 were unstable, and therefore site S5 was not included in the Miocene pole calculation. As is shown in Figure 4, site S5 diverged markedly from the cluster of other Miocene sites.

Sites S2, S7, and S8 were found to be completely unstable and are not considered further in the report. Of 101 samples collected at the remaining nine sites, 25 were rejected as unstable (Table 1).

High stability has often been attributed to single-domain carriers of the magnetism in a rock [*Nagata*, 1965; *Strangway et al.*, 1968; *Dunlop*, 1968]. In order to determine the effective domain structure in the specimens defined as stable by the stability criteria, 14 specimens representing the spectrum of coercivities were magnetized to saturation and af demagnetized. *Lowrie and Fuller* [1971] suggest such a test for igneous rocks, involving a comparison of the saturation isothermal remanent magnetism (IRM) demagnetization curve with that of NRM, to determine the dominant domain structure in a specimen. In single-domain carriers the saturation IRM is relatively less stable than the thermal remanent magnetism (TRM) acquired in a weak field. *Lowrie and Fuller* [1971] state that, if NRM in igneous rocks is TRM produced in the earth's relatively weak magnetic field, NRM can be substituted in experiments for weak-field TRM. As is shown in Figure 3, the saturation IRM is less stable than the NRM in specimens classified as stable, indicating that the magnetism of stable specimens is dominated by grains with single-domain characteristics. In unstable specimens the reverse is true, suggesting that unstable specimens are dominated by grains with multidomain characteristics.

Table 1 and Figure 4 show site mean directions after application of the stability criteria. A mean Miocene Cascade pole was

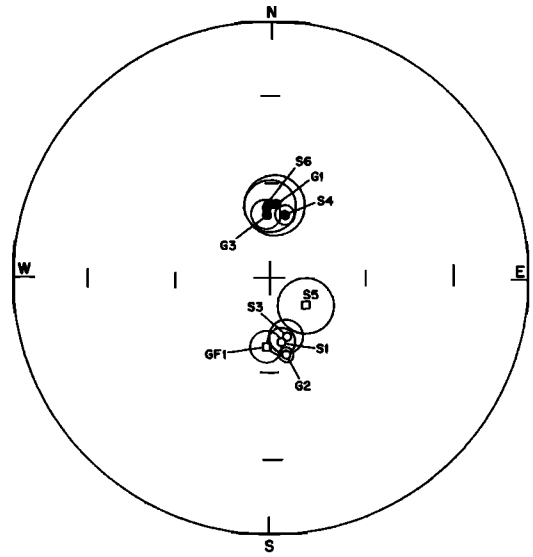


Fig. 4. Equal area plot showing site directions with their mean direction (diamond) and circles of 95% confidence. Open (solid) figures are in upper (lower) hemisphere. Circles (squares) represent directions used (not used) in the Miocene pole calculation.

obtained at 85.5°N, 208.5°E (Figure 5). The Eocene Granite Falls pluton also is shown. Its mean direction of magnetization shows no divergence from directions of intrusions dated as Miocene.

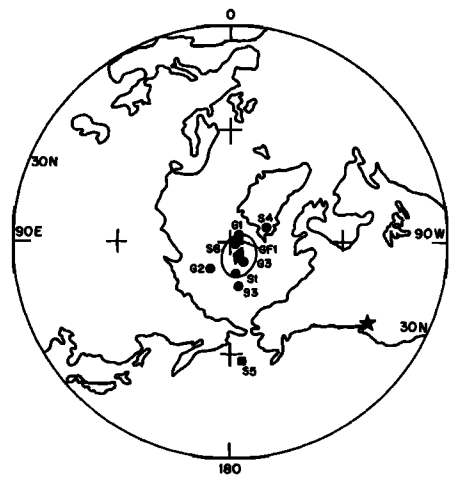


Fig. 5. Polar equal area projection showing the Miocene paleomagnetic pole calculated for the Grotto and Snoqualmie batholiths with their oval of 95% confidence. Circles (squares) represent site poles used (not used) in the mean pole calculation.

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DISCUSSION

The granitic rocks sampled in this study showed a wide variation in magnetic stability, and therefore it was necessary to analyze every specimen for stability. By using the saturation IRM experiments proposed by *Lowrie and Fuller* [1971], we found that large values of the ratio Q_d' in our samples characterize specimens with magnetism dominated by single-domain behavior, whereas specimens with low Q_d' ratios are dominated by grains with multi-domain behavior. If susceptibility χ is a measure of the amount of magnetism with very low coercivities (less than 0.45 oe) and the magnetic intensity J_d is the measure of grains with coercivities greater than the cleaning level, the ratio $J_{d/\chi}$ would indicate the relative proportion of hard and soft fractions. Thus the success of Q_d' as a stability criterion for our granitic samples from the Central Cascades may be a result of its ability to distinguish the relative proportion of high and low coercive magnetic components.

Preliminary studies of granitic rocks from the Idaho batholith indicated that the Q_d' criterion was highly reliable for distinguishing unstable samples, whereas studies of andesites from the southern San Juan Mountains in south central Colorado showed that Q_d' was not a sensitive indicator of stability for these rocks. Therefore the data available at this time would seem to indicate that the Q_d' value may not be a universal criterion for all rock types. Its principal application in this paper is that it provides a reliable semiquantitative method of testing stability in the granitic rocks from the Central Cascades that can easily be duplicated by others.

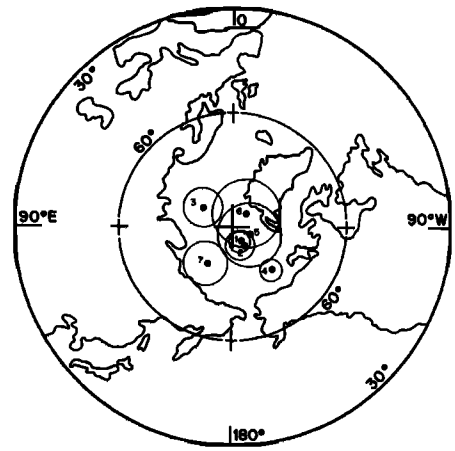


Fig. 6. Polar equal area projection comparing Miocene Cascade pole with other mid-Tertiary poles from North America. Poles are identified in Table 2.

A comparison of the paleomagnetic directions shown in Figure 4 and the location of the sampling sites within the batholith (Figure 1) indicates that there is no correlation between the distance from the walls of the batholith and magnetic polarity. This may indicate that the batholiths did not cool simply from outside to center but were formed by several separate intrusions. *Erikson* [1968] recognized separate phases in the Snoqualmie batholith as marked

TABLE 2. Data Regarding Mid-Tertiary Poles Referred to in Figure 6

Symbol	Rock Type	Area	Age, m.y.	Paleomagnetic Pole	α_{95}, deg	Reference
1	Granodiorite-diorite	Central Cascades, Washington	15 to 26	85.5°N, 208.5°E	3.2	This paper
2	Gabbro plugs	Cariboo region, south central British Columbia	8 to 15	84.9°N, 213.3°E	4.8	*Symons* [1969]
3	Basalts	Oregon at 42.6°N, 120.2°W	15	80.6°N, 56.1°E	5.3	*Watkins* [1965]
4	Ignimbrites	Hiko, Nevada	17.8	74.5°N, 222.6°E	3.2	*Gose* [1970]
5*	Sedimentary Ellensburg formation	Central Washington	10 to 14	85.0°N, 245.0°E	9.0	*Torreson et al.* [1949]
6*	Columbia River basalt	Central Washington	11 to 27	85.0°N, 317.0°E	9.0	*Campbell and Runcorn* [1956]
7	67 lava flows and welded tuffs	Utah, Nevada, Oregon	22 to 37	78.0°N, 146.0°E	5.8	*Grommé and McKee* [1971]

*NRM directions without af demagnetization.

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by lithologic changes and suggested that there may be multiple intrusions in the granodiorite phase that are not detectable petrologically.

The low dispersion of site poles shown in Figure 5 suggests that there has been little, if any, post-Miocene tectonic disturbance within the batholiths. Furthermore, the proximity of the Cascade pole to the present geographic pole and other mid-Tertiary poles suggests that there has been no discernable translation or rotation of the area relative to North America since the Miocene (Figure 6 and Table 2). Therefore our data suggest that any major tectonic movements postulated for the Cascades must be pre-Miocene. Preliminary data from the Granite Falls stock (Figure 4 and Table 1) dated at 43 ± 0.7 m.y. [*Yeats*, 1970] are similar to the Miocene data and suggest that this conclusion may hold for the post-Eocene as well.

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